

Develop a research question

Before we develop the survey let's start with the bigger question this evaluation is designed to answer.

I'll double check my notes, but I don't think we have one of those.



Week 11			
M 10/28	The Lab Replication Project (LRP): Introduction and topic selection; Literature review and hypothesis generation (brainstorm ideas)	22-23	Idea proposal due by midnight
W 10/30	The LRP: Literature review and hypothesis generation (discuss the literature review)	22	Introduction ideas due by midnight
Week 12			
M 11/4	The LRP: Study design and sampling plan	23	Study design due by midnight
W 11/6	<i>No Class (Instructor out of town)</i>		
Week 13			
M 11/11	The LRP: Variables and measurement	24	Measures due by midnight
W 11/13	The LRP: Analysis plan	25	Analysis plan due by midnight
Week 14			
M 11/18	OSF preregistration		Preregistration due by midnight
W 11/20	Group workday for presentations		Presentation slides due by midnight

Timeline

How to work together effectively?

- **Four roles**
 - **Facilitator:** Lead the discussion, ensure everyone has a chance to speak and keep the conversation focused.
 - **Timekeeper:** Make sure the group stays on track and finish assigned tasks within the allocated time, keep things moving if the discussion gets sidetracked or stuck.
 - **Notetaker:** Takes notes during the discussion, write down key points, decisions, and any tasks that need to be completed.
 - **Critical Thinker:** Challenge ideas, offer alternative perspectives, so that the group can think critically about every decision.

What to submit?

- **Group notebook**

- Keep everything in the same shared document (e.g., Google doc, OneDrive doc)
- One person **submit the assigned portion** of the day
- Need to note who's taking each role for every submission (must rotate)
- Note meeting date and time, topic, who's present, and who's absent

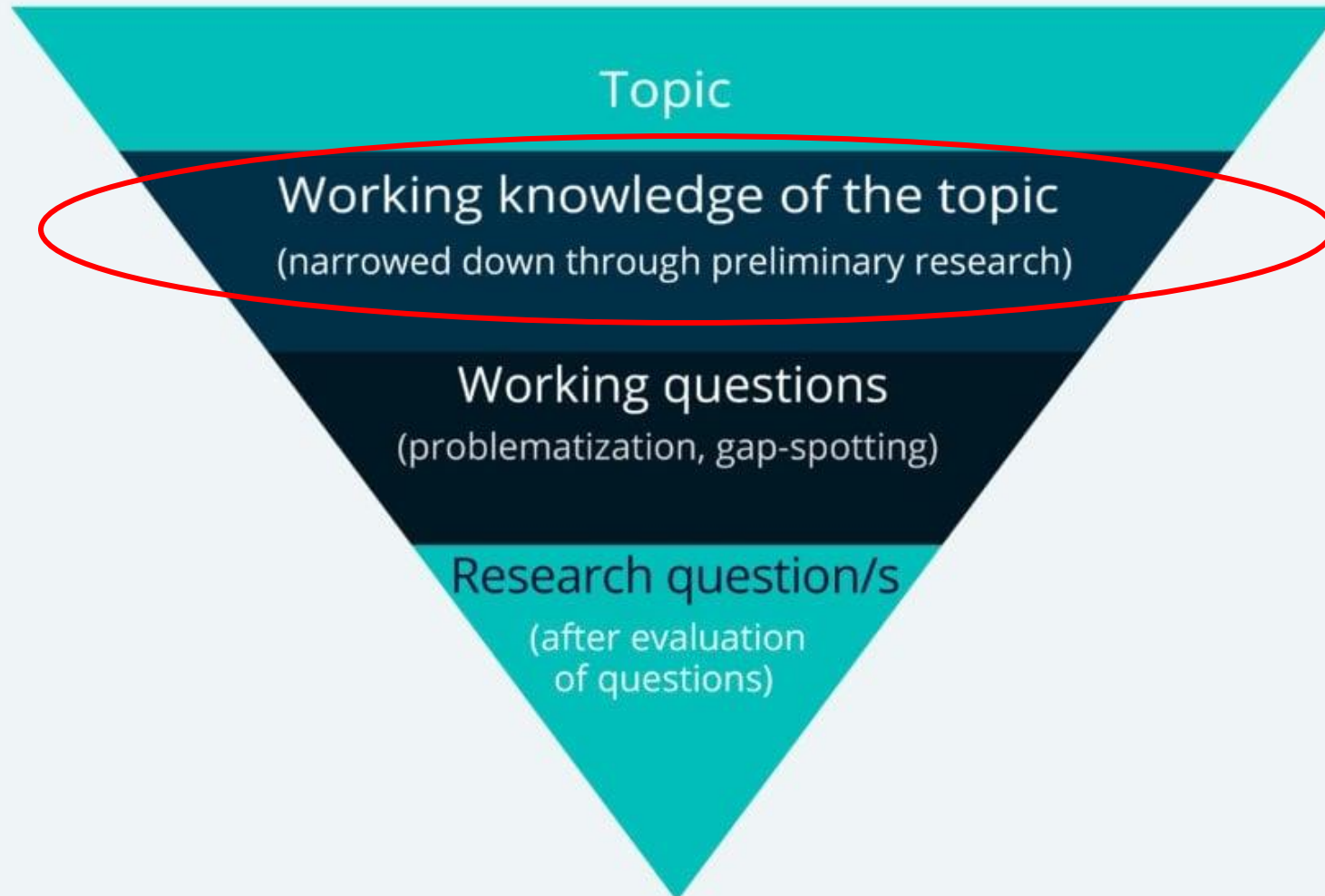
What if I need to miss a class?

- Communicate with your group member as soon as possible.
- Your group might find another time to meet and complete the group notebook **before** the next class period.
 - Need to ask for extension (GN typically due M/W midnight)
- Retrospectively... add additional useful notes/resources/comments to your GN to receive your GN points.

How to develop a research question?

1. Choose your topic (you already did)
2. Gather background information
 - Review paper
 - Textbooks

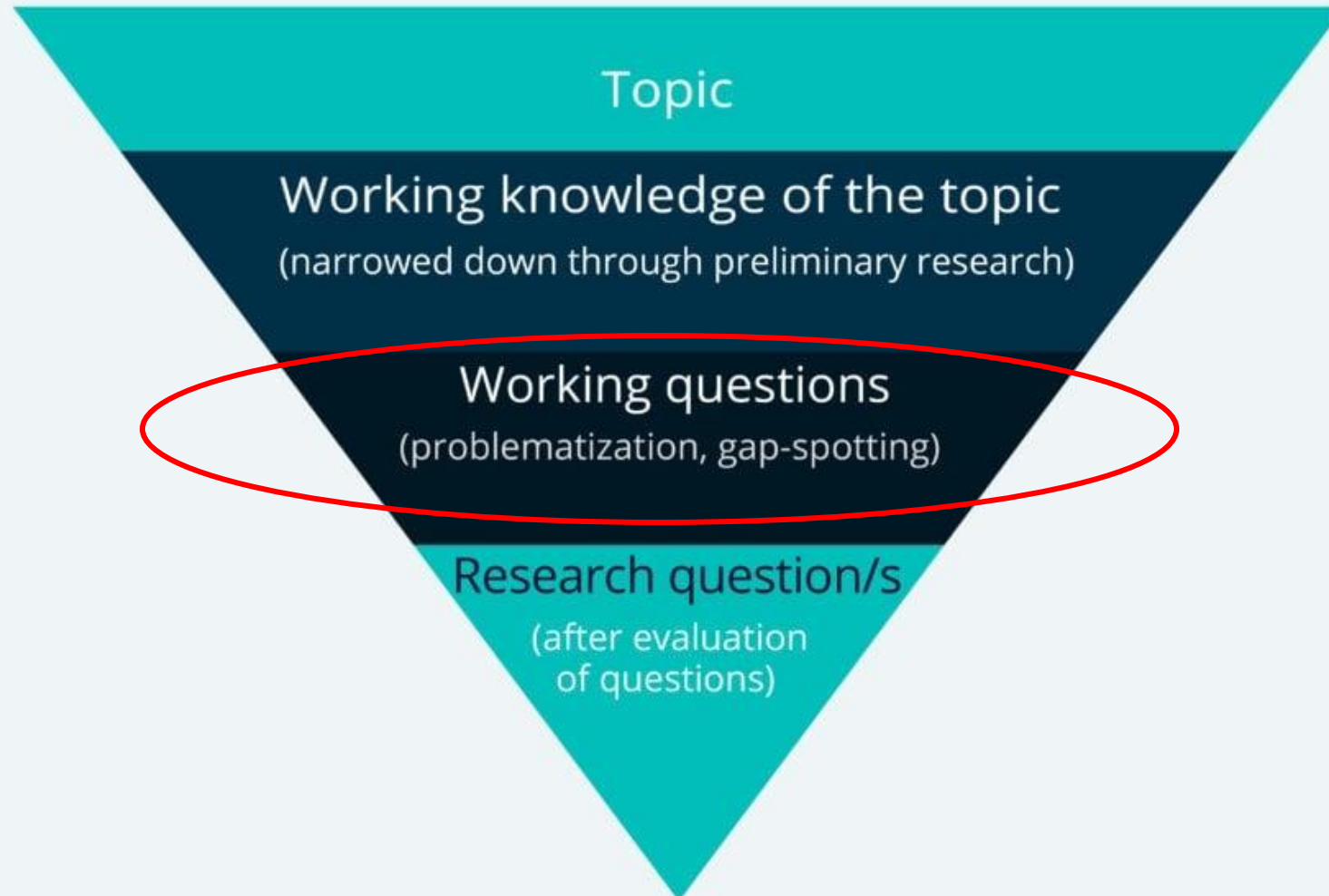
The Path to a Research Question: From Broad Topic to a Specific Question



How to develop a research question?

1. Choose your topic (you already did)
2. Gather background information
 1. Review paper
 2. Textbooks
3. What subtopics relate to the broader topic?
4. What questions do these sources raise? → GAP

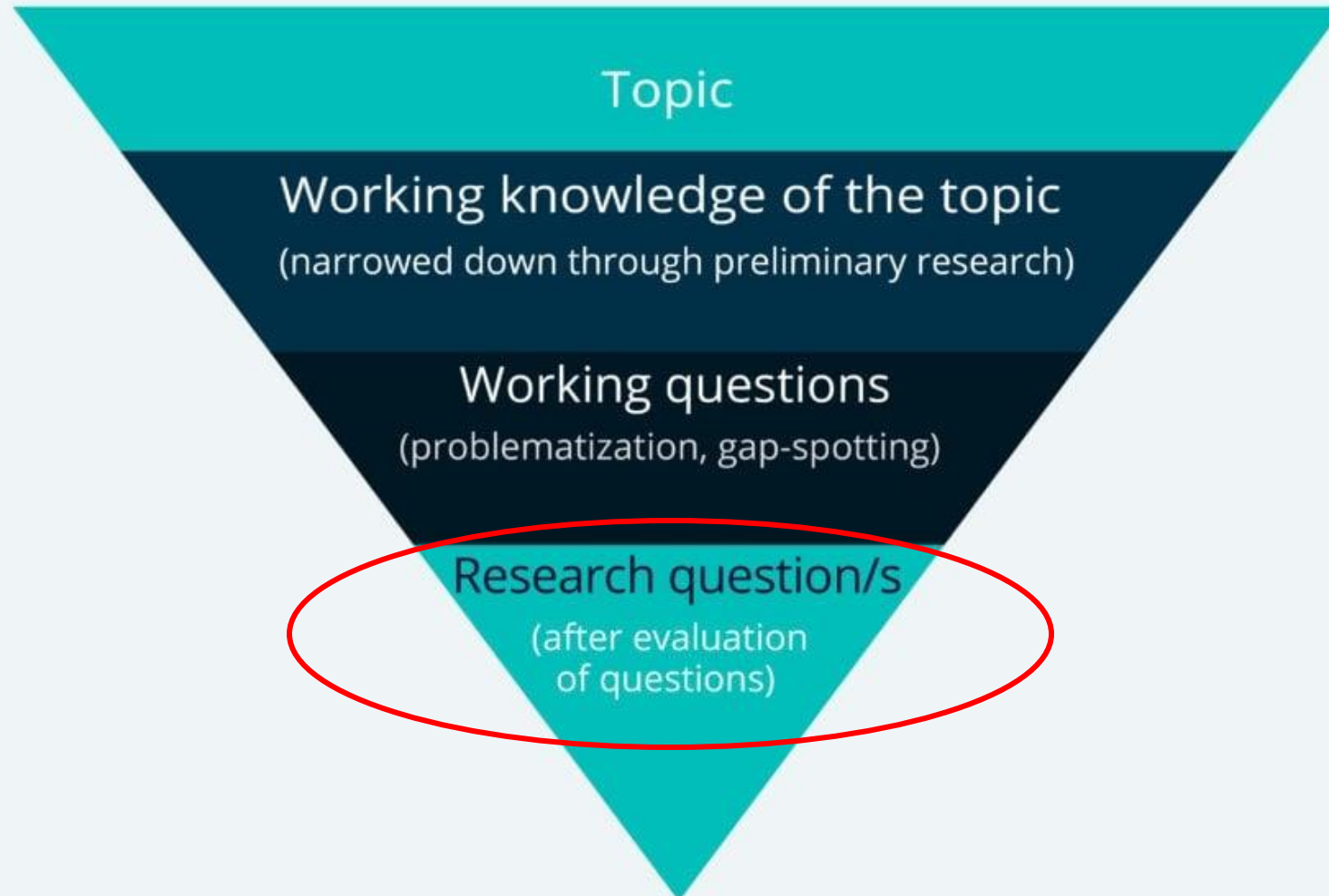
The Path to a Research Question: From Broad Topic to a Specific Question



How to develop a research question?

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2. Gather background information
 1. Review paper
 2. Textbooks
3. What subtopics relate to the broader topic?
4. What questions do these sources raise? → GAP
5. What do you find interesting about the topic?
6. Who would be interested in this issue?
7. What are some implication of your potential research question?

The Path to a Research Question: From Broad Topic to a Specific Question



From background knowledge to research question

- Explore questions
 - “How?” and “Why?”
 - “So what?”
 - Identify multiple questions you find engaging, and which could be explored further through research
- Determine and evaluate your research question
 - Is it clear and focused? Is it testable? Is it too simple or too complex?
- Hypothesize (on Wednesday)
 - Prediction (educated guess) of the results

From background knowledge to research question

- Clarity

- Unclear: Why are social networking sites harmful?
- Clear: How are online users experiencing or addressing privacy issues on social networking sites like Instagram and Facebook?

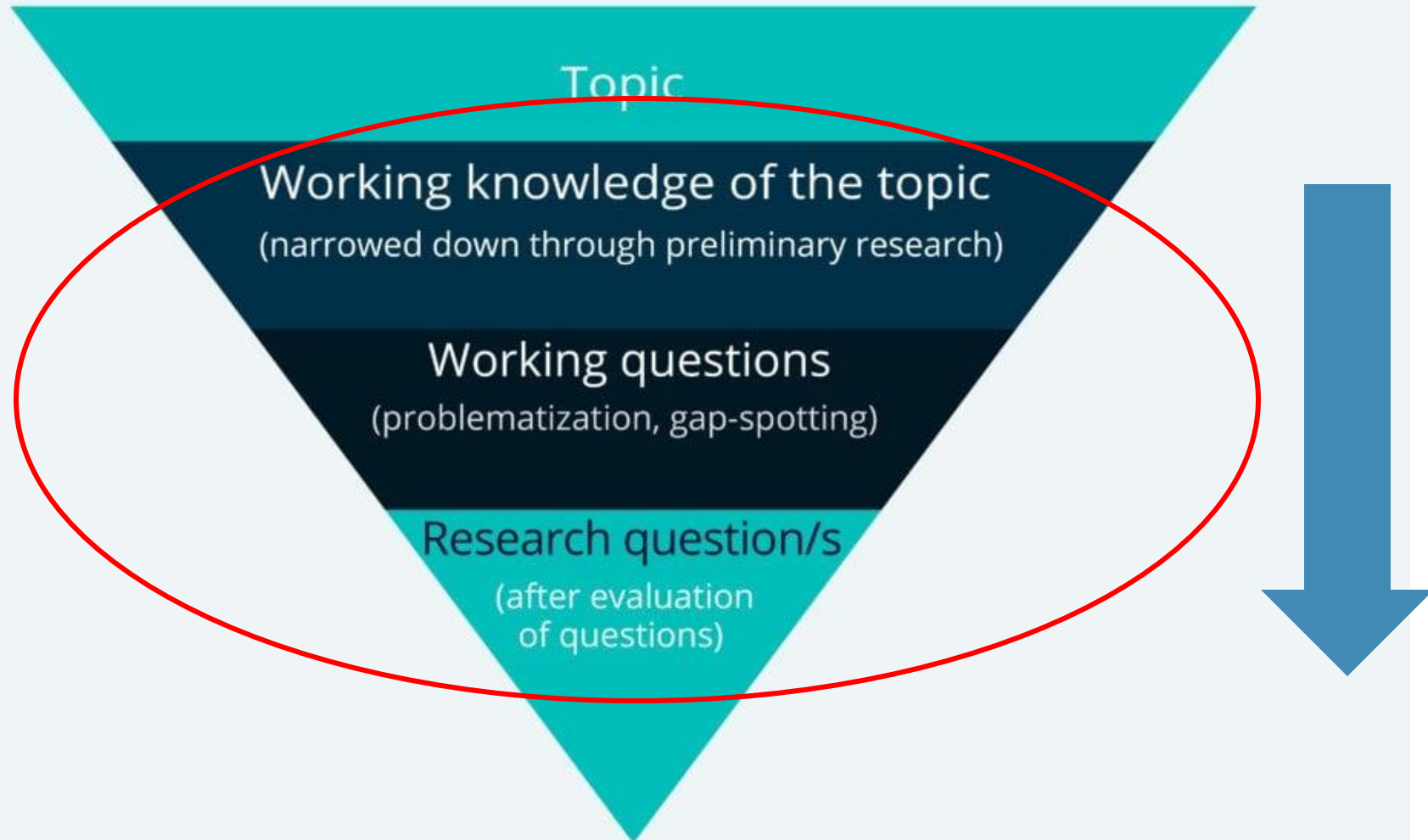
- Focused

- Unfocused: What is the effect on the environment from global warming?
- Focused: How is glacial melting affecting penguins in Antarctica?

- Complexity

- Too simple: How are doctors addressing diabetes in the U.S.?
- Appropriately complex: What are common traits of those suffering from diabetes in America, and how can these commonalities be used to aid the medical community in prevention of the disease?

The Path to a Research Question: From Broad Topic to a Specific Question



Today's Group Notebook

- Describe the working knowledge of the topic (don't forget to cite sources)
- Propose at least three working questions
 - Make sure each question is clear, focused, and appropriately complex
- Decide one final research question that is testable, most interested by the group, with good implication (can potentially solve some real-world problems).
 - Keep in mind that your final project needs to be an experiment, and your main outcome variable will need to be a continuous variable